Masaki Imagawa

Curriculum Vitae Last updated: October 2026

Contact Information

Institute of Mathematics, Academia Sinica
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Academic Appointments

Postdoctoral Fellow Aug.2025–Present
Institute of Mathematics, Academia Sinica, Taipei, Taiwan
Mentor: Shih-Hsien Yu

Education

Ph.D. in Informatics, Kyoto University Apr.2022–Jul. 2025
Dissertation: Convergence analysis of elliptic regularization for advection equations in a bounded domain
Advisor: Hiroshi Fujiwara

Master of Informatics, Kyoto University Apr. 2020–Mar. 2022

B.Sc. in Mathematics, Kyoto University Apr. 2016–Mar. 2020

Research Interests

Partial Differential Equations; Numerical Analysis; Conservation Laws; Viscous Shock Waves; Boundary Layers

Fellowships & Awards

SPRING program research fellow Apr. 2022–Mar. 2025
Dividion of Graduate Studies, Kyoto University / Japan Science and Technology Agency (JST)

Publications

Journal Articles

Masaki Imagawa and Daisuke Kawagoe, A revisit on well-posedness of a boundary value problem of a stationary advection equation without the separation condition, *J. Math. Anal. Appl.* **551** (2025), no. 2, part 2, Art. 129695.
doi:10.1016/j.jmaa.2025.129695

Conference Proceedings

Masaki Imagawa and Yuusuke Iso, Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain, in *2025 Days on Diffraction*, IEEE, 2025, 90–95.
doi:10.1109/DD66835.2025.11263497

Preprints

Masaki Imagawa and Daisuke Kawagoe,
On strong convergence of an elliptic regularization with the Neumann boundary condition applied to a stationary advection equation, arXiv:2303.17904, 2023.

Presentations

Talks

Interaction of Viscous Shock Waves with a Boundary — Oct. 2026
Analysis Seminar, Academia Sinica — Online

Shock Waves and Boundary Layers for the Viscous Burgers Equation — Oct. 2026
National Chung Hsing University Applied Mathematics Seminar — Taichung, Taiwan

Stability and Boundary Interaction of Viscous Shock Waves for the Burgers Equation — Aug. 2026
19th EASIAM Annual Meeting — Jeju, Korea

Stability of Shock Waves for the Viscous Burgers Equation with Boundary Interactions — Apr. 2026
Taipei Postdoc Seminar, National Center for Theoretical Sciences — Taipei, Taiwan

Stability of Viscous Shock Waves for the Burgers Equation in a Half-Line — Mar. 2026
Computational Mechanics Seminar, Kyoto University — Kyoto, Japan

Asymptotic stability of traveling waves for scalar viscous conservation laws with boundary — Nov. 2025
Applied Mathematics Freshman Seminar 2025, Kyoto University — Kyoto, Japan

Convergence estimates for singular perturbation of advection equations with mixed boundary conditions — Jul. 2025
Kyoto University Applied Mathematics Seminar — Kyoto, Japan

Convergence analysis of perturbed advection equations in a bounded domain — Jul. 2025
Numerical Analysis Seminar, The University of Tokyo — Tokyo, Japan

Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain — Jun. 2025
Days on Diffraction — Online

Boundary value problems of advection equations and its regularized problems — Mar. 2025
Analysis Seminar, Academia Sinica — Online

On a Boundary Value Problem of a First-Order PDE and Its Regularization Problems — Mar. 2024
2024 Japan-Taiwan Joint Workshop on Numerical Analysis and Inverse Problems
Kyoto University — Kyoto, Japan

Convergence Rates of Elliptic/Parabolic Regularization Applied to Advection Equations with Boundary Conditions — Sep. 2023
The 5th Workshop on Recent Development of Mathematical Fluid Dynamics and Hyperbolic Conservation Laws — Jeju, Korea

On the Optimal Convergence Rate of Elliptic Regularization Methods Applied to Advection Equations — Mar. 2023
The 19th Joint Meeting of JSIAM Activity Groups, Okayama University of Science — Okayama, Japan

On Convergence Rates of an Elliptic Regularization Applied to a Stationary Advection Equation — Mar. 2023
The 13th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics
National Center for Theoretical Sciences — Online

On Convergence Estimation of Elliptic Regularization Applied to Boundary Value Problem of Advection Equations — Mar. 2022
2022 Japan-Taiwan Joint Online Workshop on Numerical Analysis and Inverse Problem — Online

Numerical Computation of Advection Equations Using Elliptic Regularization — Sep. 2021
Annual Meeting of the Japan Society for Industrial and Applied Mathematics — Online

Posters

Convergence rates of elliptic regularization applied to advection equations with boundary conditions — Dec. 2023, Taipei, Taiwan
NCTS–Kyoto Mathematics Symposium, National Center for Theoretical Sciences

On the Convergence Rate of Singularly Perturbed Solutions to the Steady Advection Equation — Feb. 2023, Kyoto, Japan
The 17th ICT Innovation, Kyoto University

Teaching Experience

Teaching Assistant, Kyoto University — Apr. 2020–Jan. 2024
Seminar on Applied Analysis I (Spring–Fall 2021)
Seminar on Applied Analysis II (Spring–Fall 2022)
Seminar on Nonlinear Mathematics (Fall 2023, Fall 2024)
Special Study Course (Mathematical Science) (Spring–Fall 2022)
Nonlinear Mathematics (Spring 2021, Spring 2022, Spring 2023)
Foundations of Numerical Computation (Spring 2021, Spring 2022, Spring 2023)
Function Theory on a Complex Variable (Fall 2021)
Advanced Calculus I (Spring 2020, Fall 2020)

Outreach

Breakout Session Instructor, Kyoto University Campus Guide — Dec. 2024
Research outreach program for high school students, organized in collaboration with the Osaka Prefectural Board of Education

Poster Session Advisor, Osaka Science Day — Oct. 2024
Provided feedback on high school students' inquiry-based research presentations.

Outreach Lecturer, Kyoto University Learning Coordinator Initiative
Developed and delivered outreach lectures on mathematical modeling to high school students.
- Tosajuku Senior High School — Dec. 2024
- Fukui Prefectural Fujishima High School — Nov. 2024
- Shanghai Japanese School Senior High School (Online) — Nov. 2023

Student Research Mentor, Kyoto Municipal Horikawa High School — Apr. 2023–Sep. 2024
Mentored high school students in inquiry-based research projects.

Qualifications

Specialized Teaching Certificates in Mathematics for Lower and Upper Secondary Schools, Japan — Mar. 2022